

LESSON 7.2

SECTORS OF A CIRCLE



LESSON INTRODUCTION

MA.7.DP.1.4 Use proportional reasoning to construct, display, and interpret data in circle graphs.

LEARNING TARGETS

- I can find the percentage the area of a sector represents in a circle.
- I can find the central angle measure of a sector given the percentage of the circle a sector represents.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.9.DP.1.1 Given a set of data, select an appropriate method to represent the data depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

ESSENTIAL QUESTIONS

- How do you find the central angle measure of a sector given the percentage of the circle that the sector represents?
- How do you use the central angle measure to calculate the percentage that the area of a sector represents?

LESSON NARRATIVE

Students use a protractor to find the angle measure of sectors of a circle. Students use the angle measure to write the ratio that angle measure represents out of 360° . Finally, they convert the ratio to the percentage of the area of the circle. This lesson is intended to prepare students for constructing circle graphs in the next lesson.

COMPONENT SUMMARY

7.2.1 WARM-UP (~5 MIN.)

Justify which circle graph doesn't belong

7.2.2 GUIDED INSTRUCTION (15-20 MIN.)

Measure central angles and use them to calculate percentages of circles represented

7.2.3 GUIDED PRACTICE (10-15 MIN.)

Practice measuring central angles and calculating percentages sectors of a circle represent

7.2.4 WRAP-UP (~5 MIN.)

Demonstrate mastery of lesson learning target

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Sector

MATERIALS

- Protractors for students

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not align the side of the angle with 0° on the protractor when measuring angles.
- Students may inconsistently use either the top or the bottom scale on the protractor, resulting in inaccurate measures.
- Students may confuse the percentages and angle measurements of the sectors. For example, a sector that represents 50% of a circle does not have an angle measure of 50° .

THINGS TO CONSIDER

- 7.2.2 Guided Instruction asks students to measure angles using a protractor. Model how to use a protractor, highlighting the importance of placing the midpoint of the protractor on the vertex of the angle, lining up one side of the angle with the zero line on the protractor, and reading the angle measurement where the other side crosses the number scale.

LITERACY AND LANGUAGE ROUTINES

Which One Doesn't Belong

In 7.2.1 Warm-Up, students decide which of four circle graphs doesn't belong. This routine has a low floor and a high ceiling. There are no wrong answers as long as the student justifies their selection. After identifying one answer, challenge students to think of additional reasons each circle graph doesn't belong. Students compare the attributes of circle graphs and practice using vocabulary terms.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

In 7.2.2 Guided Instruction and 7.2.3 Guided Practice, students have repeated opportunities to practice determining the percentages and angle measurements that represent sectors. Consider utilizing the suggested problems to help students recognize that equivalent fractions can be an efficient method of finding angle measurements and percentages.

DIFFERENTIATE INSTRUCTION

Students are provided with a kinesthetic entry point as they measure the central angles of circles in 7.2.2 Guided Instruction. Students are also provided with a visual entry point by seeing the circle graphs. Moreover, in the 7.2.1 Warm-Up discussion, students are provided with an auditory entry point. In 7.2.3 Guided Practice, if students are struggling to determine the central angle measurements or percentages, consider displaying an anchor chart model to reference when setting up their equivalent fractions.

$$\frac{\text{Part}}{100} = \frac{\text{angle measure}}{360}$$

Additional differentiation opportunities are denoted throughout the lesson guide.

7.2.1 WARM-UP: WHICH ONE DOESN'T BELONG?

TEACHER MOVES

Which One Doesn't Belong

Students decide why one out of four circle graphs doesn't belong. This routine has a low floor and a high ceiling. There are no wrong answers as long as the student justifies their selection with reasoning. After identifying one answer, challenge students to think of additional reasons each circle graph doesn't belong. Students compare the attributes of circle graphs and practice using vocabulary terms.

Consider polling students to determine their selections before identifying a couple of students to share out loud, if time permits.

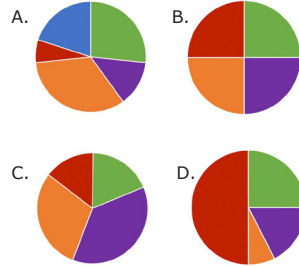
DIFFERENTIATE INSTRUCTION

Rather than having students raise their hands, have students hold their hands over their hearts and lift fingers to show how many solutions they have found. Students who work slower can feel discouraged and stop working when they see other students raising their hands.



7.2.1 Warm-Up: Which One Doesn't Belong?

- With your partner, review the four circle graphs and determine which one doesn't belong. Explain your choice.

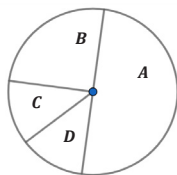


Answers will vary. Circle B does not belong because it is the only circle divided into equal-sized sectors. Circle C does not belong because it is the only circle that has a purple sector which takes up more than $\frac{1}{4}$ of the circle graph.



7.2.2 Guided Instruction: Central Angle Measures of Sectors

1. The circle below is divided into sectors.



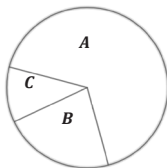
- a. Use your protractor to measure the angle for each sector. Then, complete the table.

Sector	Central Angle Measure in Degrees	Ratio of Angle Measure to 360°	Percentage of the Circle
A	180°	$\frac{180}{360}$	50%
B	90°	$\frac{90}{360}$	25%
C	45°	$\frac{45}{360}$	12.5%
D	45°	$\frac{45}{360}$	12.5%

- b. Discuss with your partner how the central angle is related to the percentage of the circle. Summarize your discussion.

The ratio of the central to the total degrees in a circle, 360, represents the fraction of the circle in that sector. Multiply the fraction by 100 to find the percentage of the circle. The central angle is the percentage of 360° represented by that sector.

2. A circle with three noncongruent sectors is shown.



- a. Use your protractor to measure the angle for each sector. Then, complete the table. Round your answers to the nearest whole number.

Sector	Central Angle Measure in Degrees	Ratio of Angle Measure to 360°	Percentage of the Circle
A	240°	$\frac{240}{360}$	67%
B	80°	$\frac{80}{360}$	22%
C	40°	$\frac{40}{360}$	11%
Total	360°	$\frac{360}{360}$ or 1	100%

7.2.2 GUIDED INSTRUCTION: CENTRAL ANGLE MEASURES OF SECTORS

TEACHER MOVES

Before using a protractor, develop students' reasoning and sense-making skills by asking them to estimate the degree measurements of the sectors. Then have students check the accuracy of their estimates using protractors. Finally, ask students to reflect on whether their estimates were reasonable. Consider modeling how to measure an angle with a protractor, as some students may not have done this since elementary school.

TEACHER MOVES

Model all or part of question 1 for students using a Think Aloud to talk through how to measure the central angle and use it to complete the table. Students are developing this understanding and should not be expected to do it completely on their own in this component.

TEACHER MOVES

Consider allowing students to work with a partner on question 2 if they have demonstrated readiness after question 1. If not, consider modeling question 2 using a Think Aloud and answering any student questions before they move to the 7.2.3 Guided Practice in the next component.

ENRICHMENT

Consider asking students the following:

- Is Sector A greater than or less than 180°?
- How could you use the measures of the central angles for Sectors B and C to determine the measure of Sector A?

7.2.2 GUIDED INSTRUCTION: CENTRAL ANGLE MEASURES OF SECTORS (CONTINUED)

ENRICHMENT

Consider asking students the following:

- What does 360° represent in a circle?
- How does this relate to the values 1 and 100%?

- b. What do you notice about the sums of the “total” row?

They equal one whole circle. The degrees add up to 360 which is the number of degrees in an entire circle. The ratios added together equal 1. The percentages added together equal 100%.

- c. Discuss with your partner why the total percentage may be slightly more or slightly less than 100%.

When dividing fractions, sometimes we get a repeating or a non-terminating decimal which needs to be rounded which can cause the sum to be more than 100%.

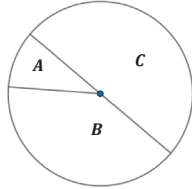
3. The table contains information for a circle with five different-sized sectors. Complete the table. Round your answers to the nearest whole number.

Sector	Percentage of the Circle	Ratio of Angle Measure to 360°	Central Angle Measure in Degrees
A	22%	$\frac{79}{360}$	79°
B	15%	$\frac{54}{360}$	54°
C	41%	$\frac{148}{360}$	148°
D	2%	$\frac{7}{360}$	7°
E	20%	$\frac{72}{360}$	72°
Total	100%	$\frac{360}{360}$ or 1	360°



7.2.3 Guided Practice: Central Angle Measures

1. A circle with three noncongruent sectors is shown.



Use your protractor to measure the angle for each sector. Then, complete the table. Round your answers to the nearest whole number.

Fraction of the Circle	Central Angle Measure in Degrees	Ratio of Angle Measure to 360°	Percentage of the Circle
A	40°	$\frac{40}{360}$	11%
B	140°	$\frac{140}{360}$	39%
C	180°	$\frac{180}{360}$	50%

7.2.3 GUIDED PRACTICE: CENTRAL ANGLE MEASURES

TEACHER MOVES

Prompt students to verify the reasonableness of their answers by finding the totals of each column in the table.

7.2.3 GUIDED PRACTICE: CENTRAL ANGLE MEASURES (CONTINUED)

DIFFERENTIATE INSTRUCTION

If students are struggling to determine the central angle measurements or percentages, consider displaying an anchor chart model to reference when setting up their equivalent

fractions. $\frac{\text{Part}}{100} = \frac{\text{angle measure}}{360}$

2. The table contains information for a circle with four different size sectors. Complete the table. Round your answers to the nearest whole number.

Sector	Percentage of the Circle	Ratio of Angle Measure to 360°	Central Angle Measure in Degrees
A	36%	$\frac{129}{360}$	129°
B	24%	$\frac{86}{360}$	86°
C	17%	$\frac{61}{360}$	61°
D	23%	$\frac{83}{360}$	83°

**7.2.4 Wrap-Up: Cool Down**

1. Determine the measure of a central angle if the percentage of the circle is 63%. Round your answer to the nearest whole number.

227°

2. What is the percentage of a circle if the central angle measures 40°? Round your answer to the nearest whole.

11%

7.2.4 WRAP-UP: COOL DOWN**QUESTIONING**

Consider our online resources for additional enrichment opportunities for students to practice or test their abilities.